

Speech Processing

Dr. Shaik Mohammad Rafi B
UG3 S2026
Indian Institute of Information Technology Sri City

Review of Signal Processing: LTI Systems

- A **Linear Time-Invariant (LTI)** system is completely characterized by its **impulse response** = response of the system to $\delta[n]$.
- Response of LTI system for any arbitrary input is

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

- Discrete time Fourier Transform of a signal is given as $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

$$\omega = 2\pi \frac{F}{F_s} \quad -\pi \leq \omega \leq \pi$$

- ω in rad/sample; $\omega=0$ to π is considered as general frequency range (ignoring negative values $-\pi$ to 0).
- F_s : sampling rate in samples/sec or Hz
- F : continuous time frequency in cycles/sec or Hz

Frequency response

The **frequency response of LTI system** is the DTFT of the impulse response:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$$

$|H(e^{j\omega})|$ (Magnitude response)
 $\angle H(e^{j\omega})$ (Phase response)

Input-output relation: $y[n] = x[n] * h[n]$

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$$

DTFT properties

Property	Time-Domain Operation	DTFT Relation
1. Delay (Time Shift)	$y[n] = x[n - n_0]$	$Y(e^{j\omega}) = e^{-j\omega n_0} X(e^{j\omega})$
2. Convolution	$y[n] = x[n] * h[n]$	$Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$
3. Symmetry (Time Reversal / Conjugation)	$y[n] = x[-n]$	$Y(e^{j\omega}) = X(e^{-j\omega})$
	$y[n] = x^*[n]$	$Y(e^{j\omega}) = X^*(e^{-j\omega})$

System function

System function $H(z)$ standard form

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

- **Zeros**: roots of the numerator
- **Poles**: roots of the denominator
- Poles and zeros are plotted in complex z -plane. $z = r^* e^{j\omega}$
- $|z|=r=1$ is called **unit circle** in z -plane

Frequency response from $H(z)$

$$H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}}$$

- **Zeros near unit circle** → spectral **nulls (valleys)**
- **Poles near unit circle** → spectral **peaks (resonances)**

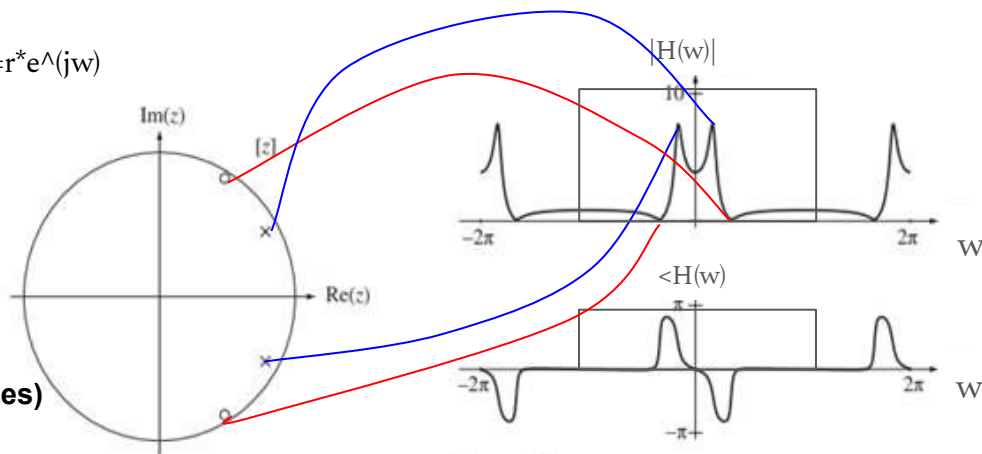


Figure 9.16

Pole-zero diagram of the transfer function,

$$H(z) = \frac{z^2 - 0.96z + 0.9028}{z^2 - 1.56z + 0.8109}$$

Figure 9.17

Magnitude and phase frequency response of the system whose transfer

$$\text{function is } H(z) = \frac{z^2 - 0.96z + 0.9028}{z^2 - 1.56z + 0.8109}$$

Autocorrelation of a signal

Auto correlation is similarity of a signal with its delayed-self.

Its a function of delay: $R_{xx}[k] = \sum_{n=-\infty}^{\infty} x[n] x^*[n - k]$

$R_{xx}[0] = \sum_{n=-\infty}^{\infty} |x[n]|^2$ gives energy of the signal.

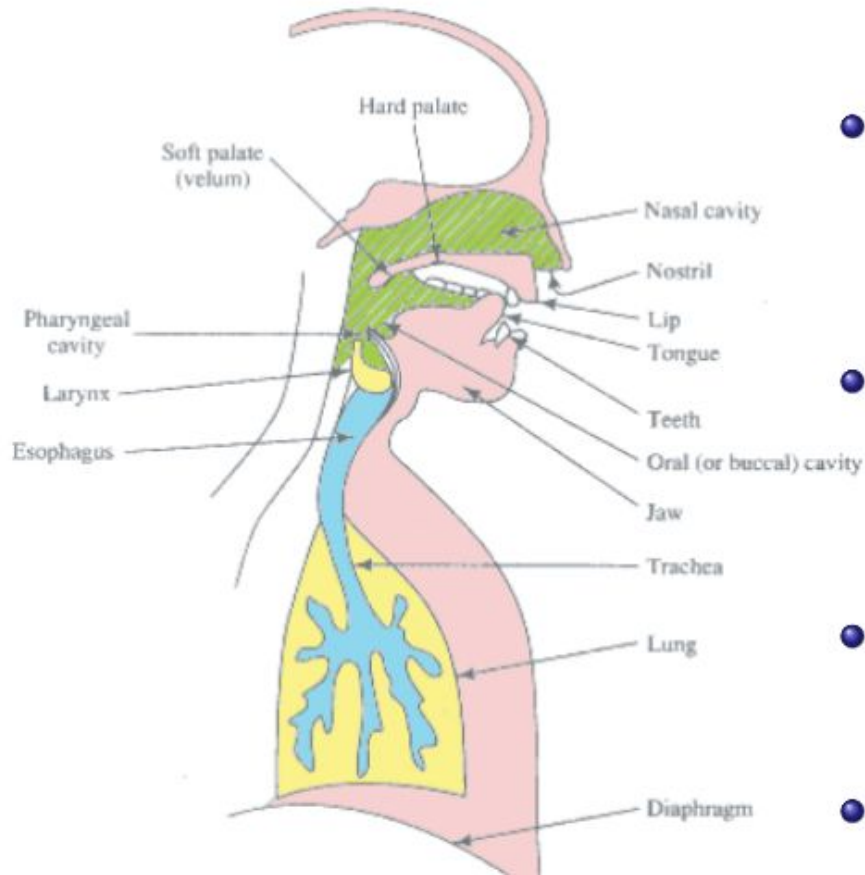
Autocorrelation is even symmetric function: $R[-k]=R[k]$.

Autocorrelation of a periodic signal is also periodic with same period.

DTFT of autocorrelation gives power spectral density.

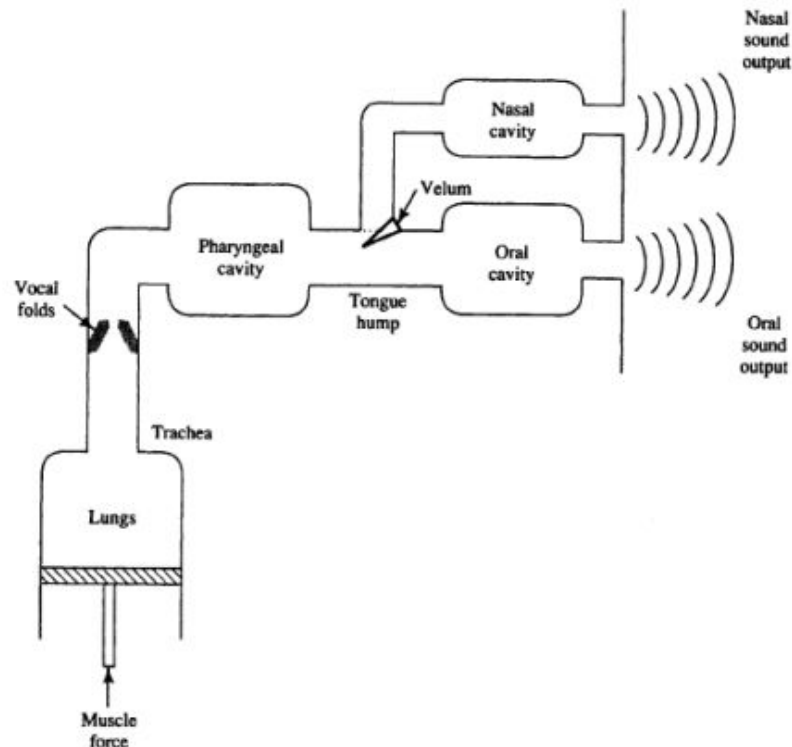
$$S_{xx}(e^{j\omega}) = \sum_{k=-\infty}^{\infty} R_{xx}[k] e^{-j\omega k}$$

Speech Production Mechanism



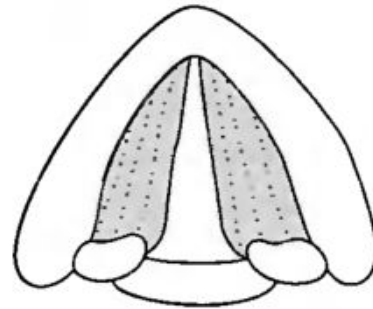
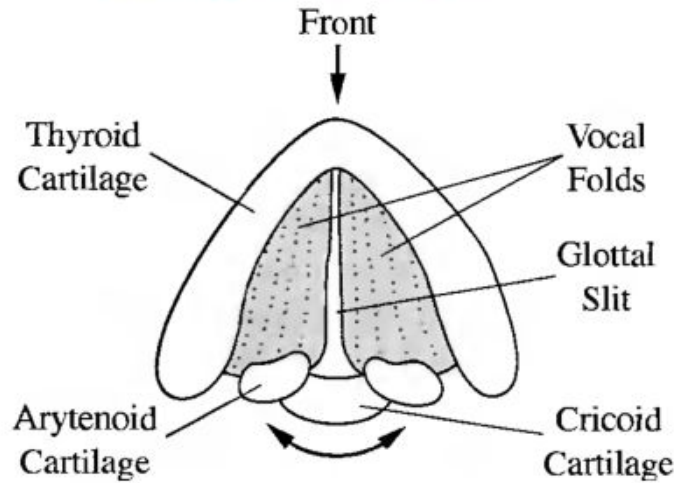
- Speech is the most sophisticated motor activity in the human body
- The motions are lightening fast and totally fluid, yet highly accurate
- Nasal cavity is much larger than oral cavity
- Animation

Block Diagram of Speech Production

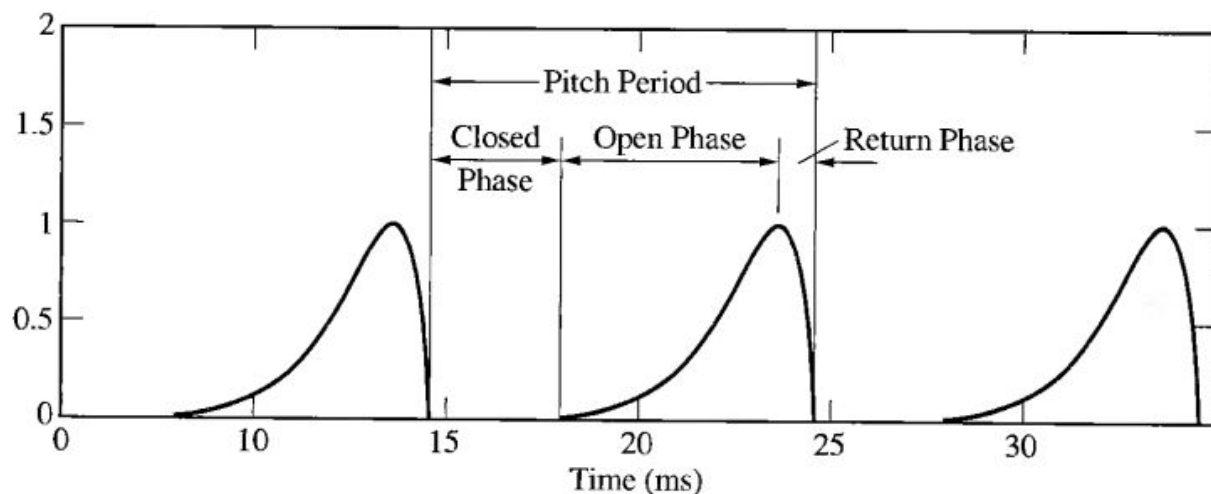


- Lungs act as source of energy
- Vocal folds chop the airflow from lungs into quasi periodic puffs of excitation
- Shape of the vocal tract determines the sound that is produced
- Velum is big enough is decouple nasal cavity, but not oral cavity

Vocal Folds - View & Operation



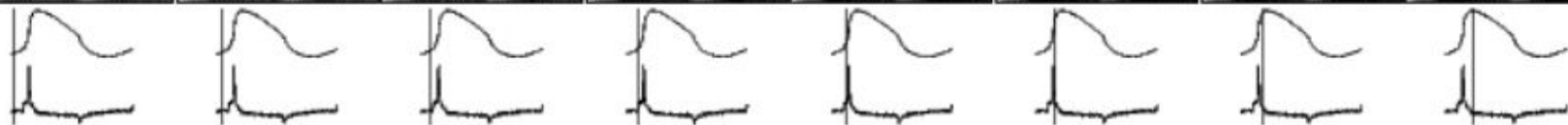
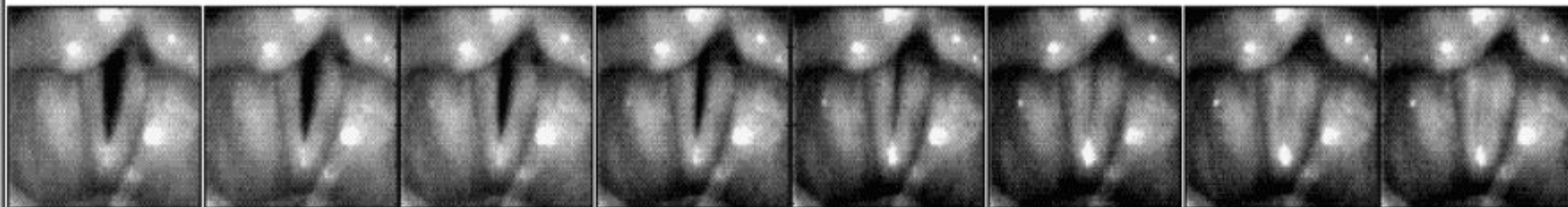
Glottal Flow



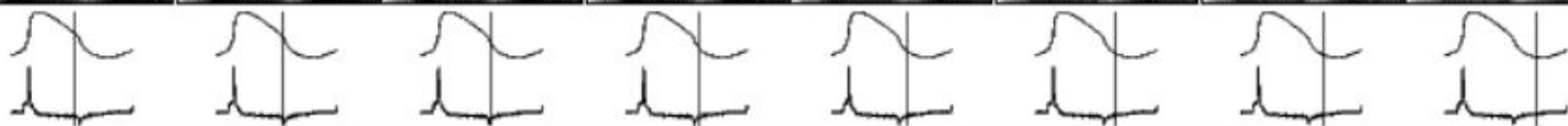
- Vocal folds are muscular tissues in the larynx
- Open when we breathe quietly and closed when we swallow
- Air coming from the lungs causes vocal folds to vibrate between open and closed positions when we talk
- Bernoulli Principle: Increase in the velocity of fluid occurs with simultaneous decrease in the pressure

Laryngeal Cycle

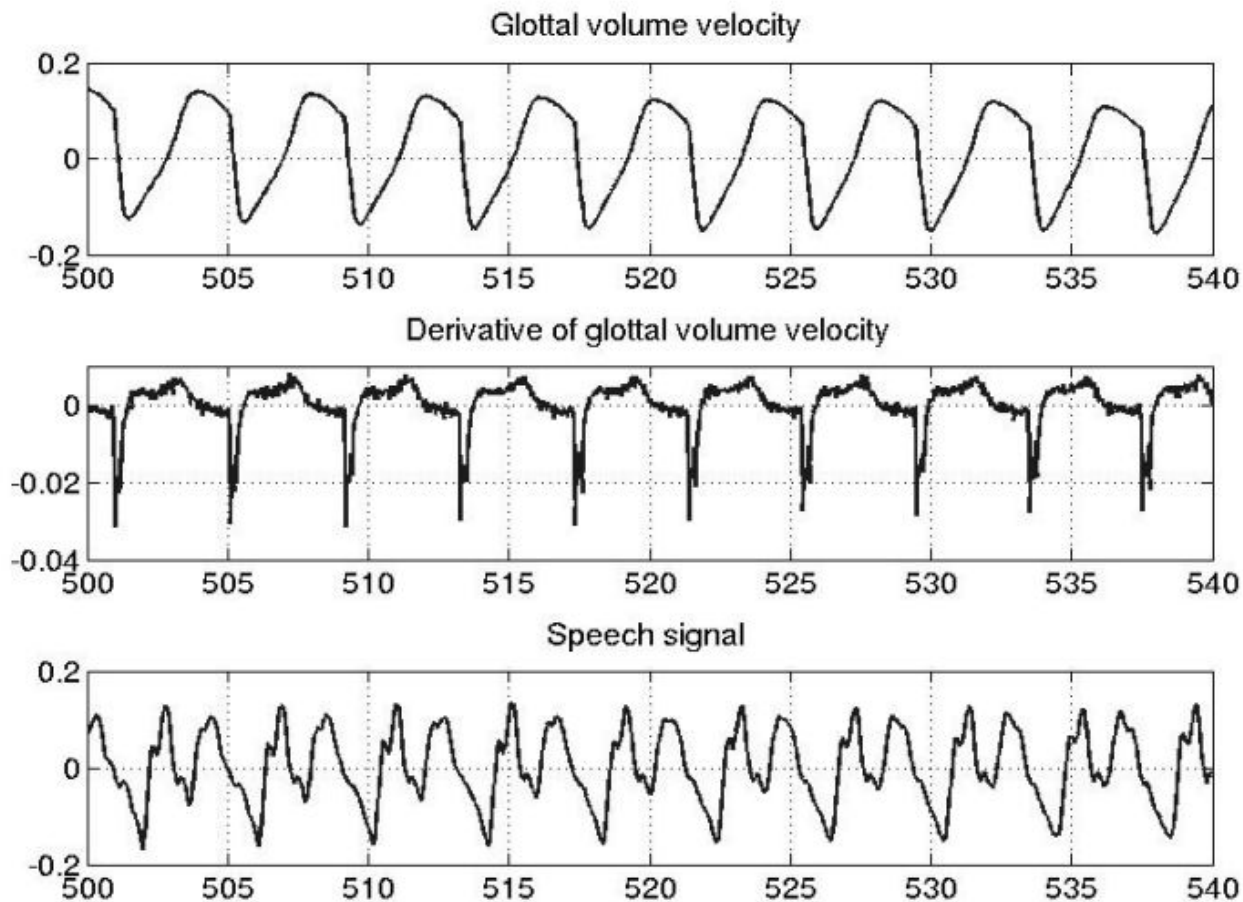
CLOSURE PHASE



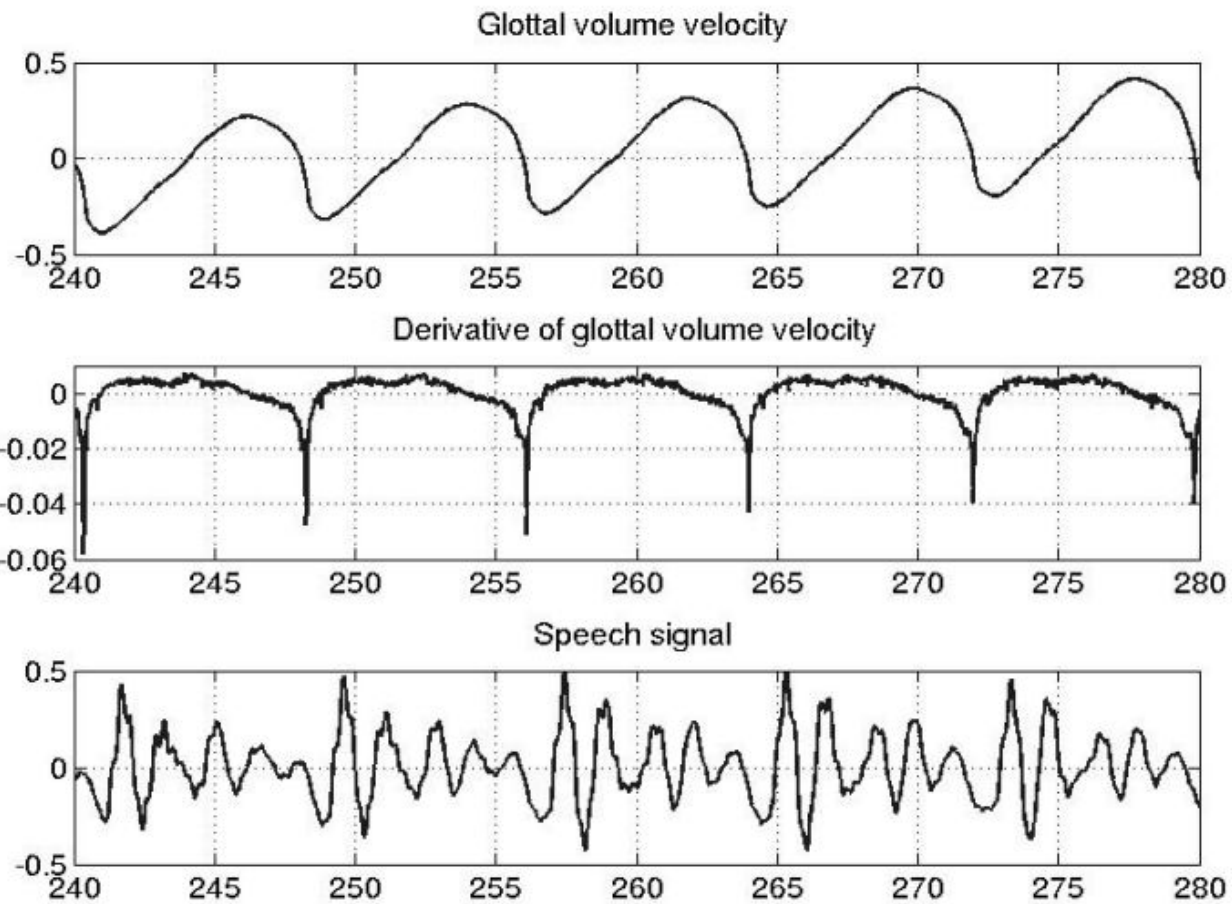
OPENING PHASE



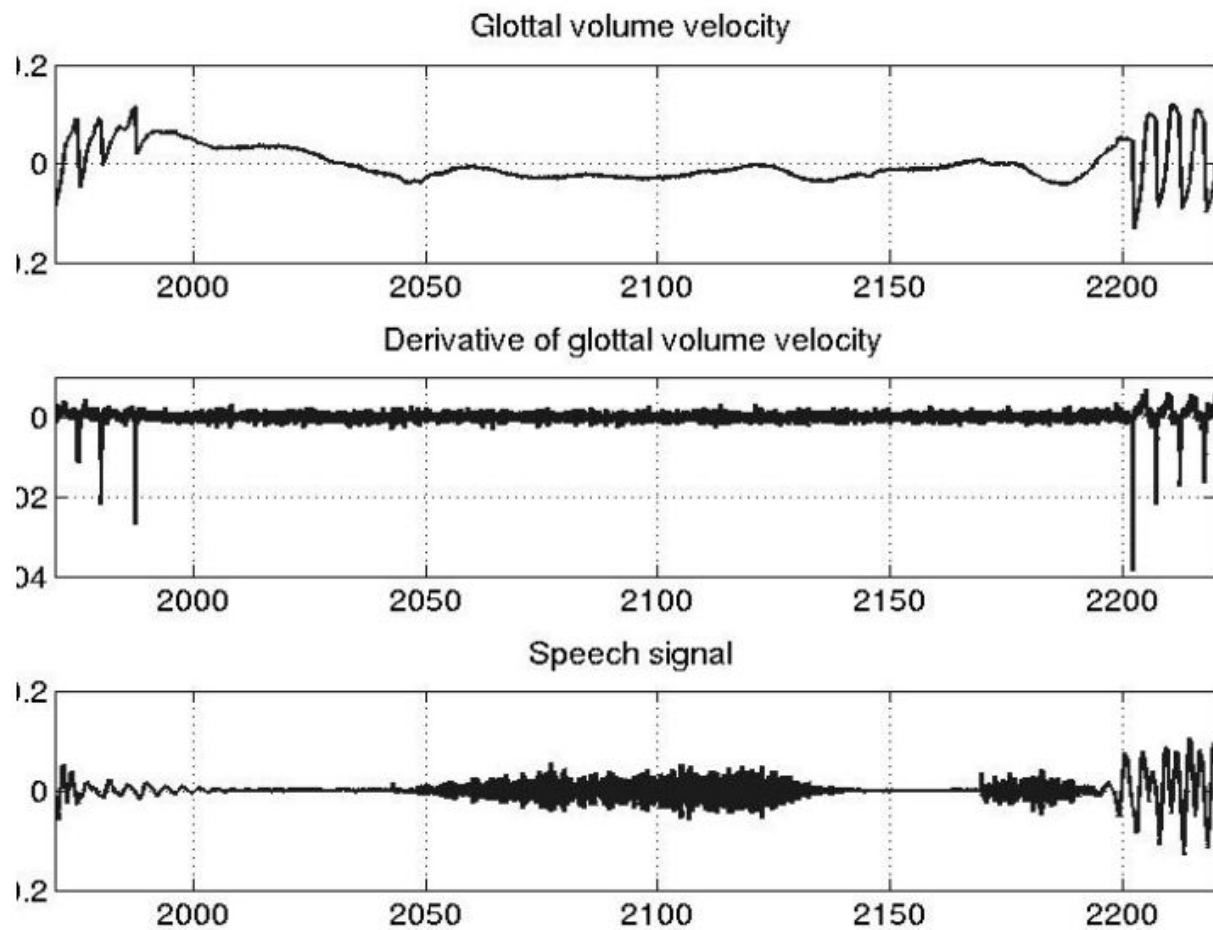
Voice Source - Female Speaker



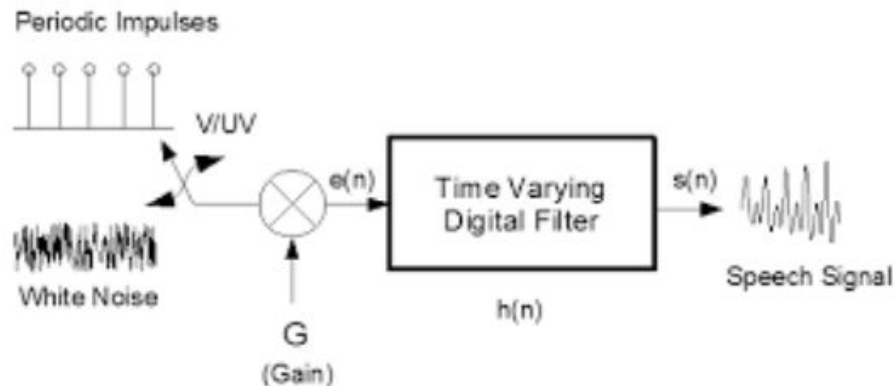
Voice Source - Male Speaker



Voice Source During Frication



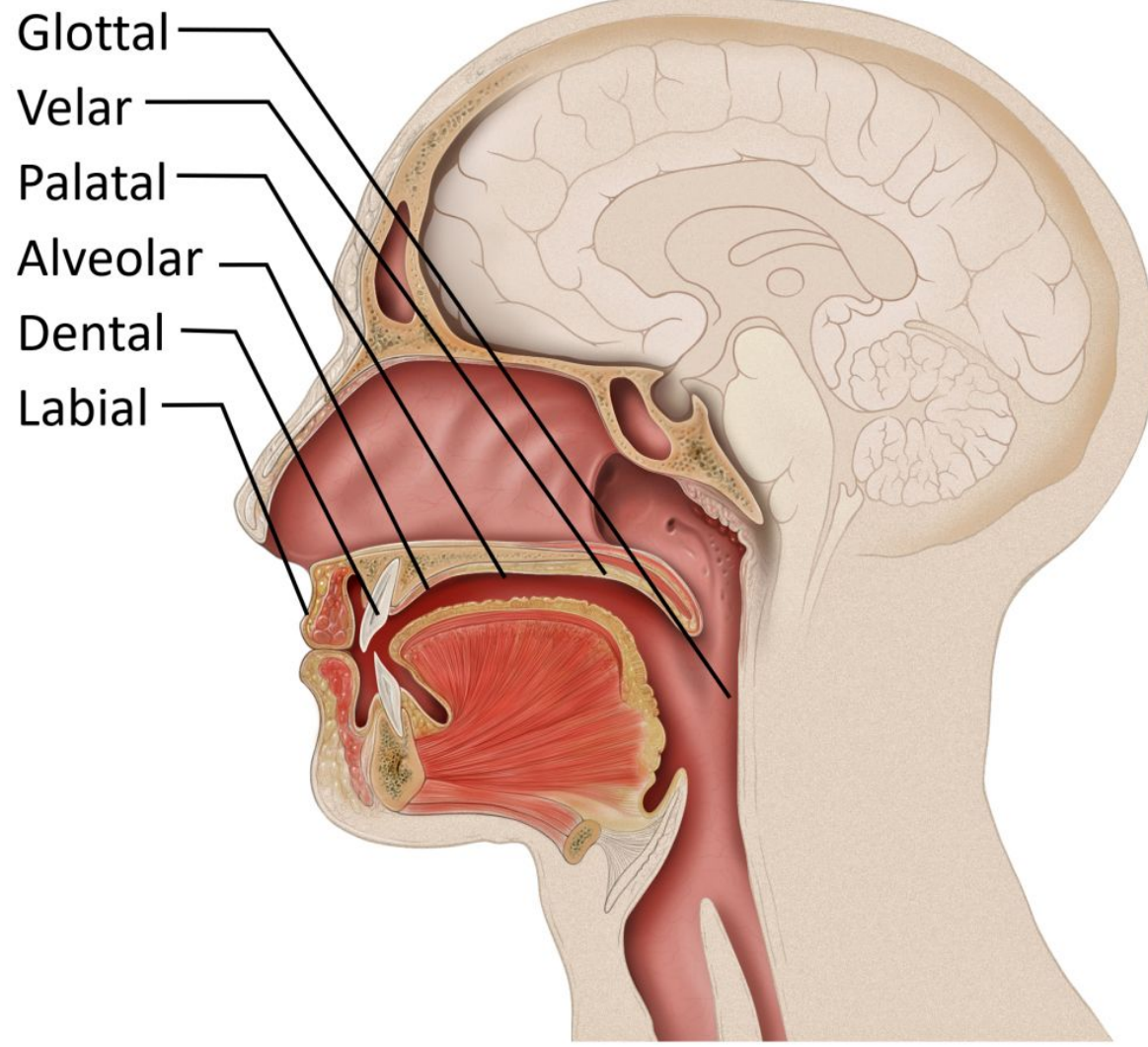
Source-Filter Model of Speech Production



- Speech is an outcome of time-varying vocal tract filter driven by a time-varying excitation
- The state of the vocal cords, the positions, shapes and sizes of the various articulators – all change slowly over time, thereby producing the desired speech sounds
- Need to determine the state of VT from waveform - inverse problem

Speech sounds classification

Manner	Description	Examples
Vowels	Sustained periodic	/i/, /e/, /a/
Plosives (Stops)	Complete closure + burst	/p, b, t, d, k, g/
Fricatives	Narrow constriction → noise	/s, z, ʃ/
Affricates	Stop + fricative	/tʃ, dʒ/
Nasals	Air through nose	/m, n, ŋ/
Liquids	Partial closure	/l, r/
Glides (Semivowels)	Vowel-like	/w, j/ (va,ya)



Place of articulation

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b		t d			ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ	n			ɳ	ɲ	ŋ	ɴ		
Trill	ʙ		r						ʀ		
Tap or Flap		ⱱ	ɾ			ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative			ɬ ɮ								
Approximant		ʋ	ɹ			ɻ	j	ɰ			
Lateral approximant			l			ɭ	ʎ	ʟ			

అ ఆ ఇ ఈ ఉ ఊ ఋ ఎ ఏ ఐ ఒ ఓ ఔ
a ā i ī u ū ṛ e ē ai o ō au
[a] [a:/æ:] [i] [i:] [u] [u:] [rɨ] [e] [e:] [aj] [o] [o:] [aw]

ka kha ga gha ṇa
[ka] [k^ha] [ga] [g^ha] [ṇa]

Palatal affricates and nasal

ca cha ja jha ña
[tʃa] [tʃʰa] [dʒa] [dʒʰa] [ɲa]

Retroflex stops and nasal

ṭa	ṭha	ḍa	ḍha	ṇa
[ṭa]	[ṭ ^h a]	[ḍa]	[ḍ ^h a]	[ṇa]

Dental stops and nasal

ta	tha	da	dha	na
[ta]	[t ^h a]	[da]	[d ^h a]	[na]

Labial stops and nasal

pa pha ba bha ma
[pa] [p^ha/fa] [ba] [b^ha] [ma]

Semivowels (liquids and glides)

ya ra la va ɭa
[ja] [ra] [la] [wa] [ɭa]

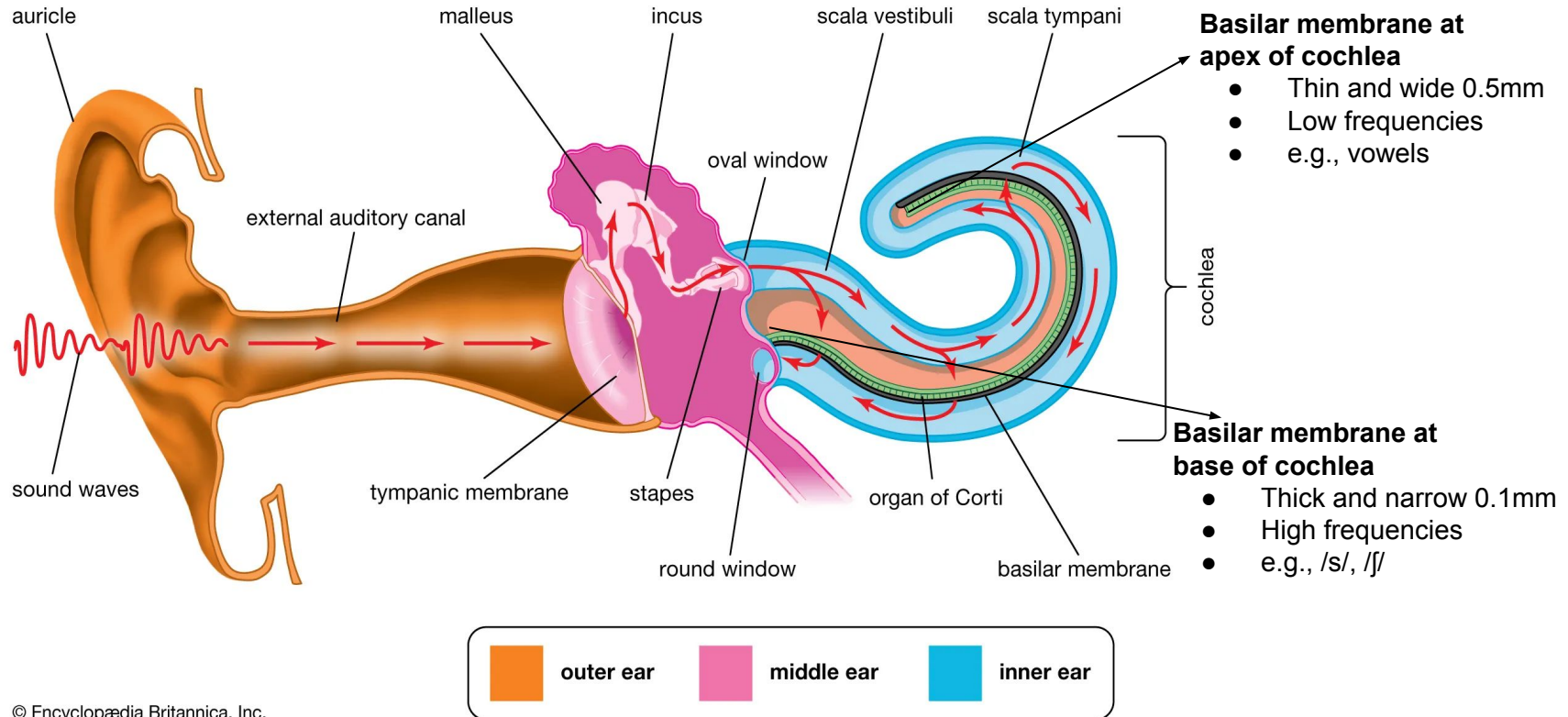
Fricatives

śa ṣa sa ha
[ʃa] [ʂa] [sa] [ha]

Audio perception by ear

engineered as **mechanical**—**hydraulic**—neural chain.

BM length is typically 32–35 mm



Basilar membrane: a tuned band-pass filter

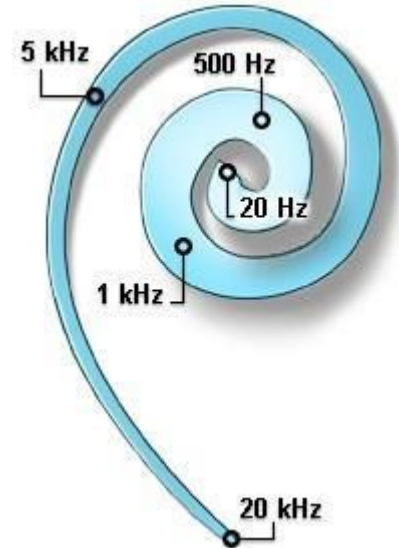
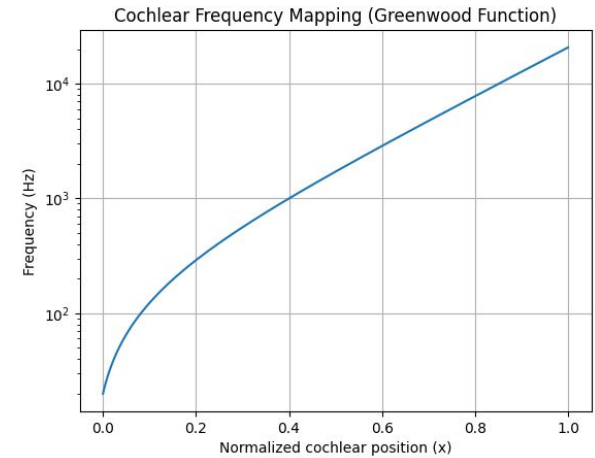
Frequency along BM: $f(x) = 165.4 (10^{2.1x} - 0.88) = 20 \text{ Hz}; x=0 \text{ (apex)}$
 $= 20,000; x=1 \text{ (base)}$

Differential of $f(x)$: $f'(x) = K * 10^{(2.1 * x)}$

Large near base $\rightarrow$ poor spatial resolution : at high frequencies

Small near apex $\rightarrow$ fine resolution: at low frequencies

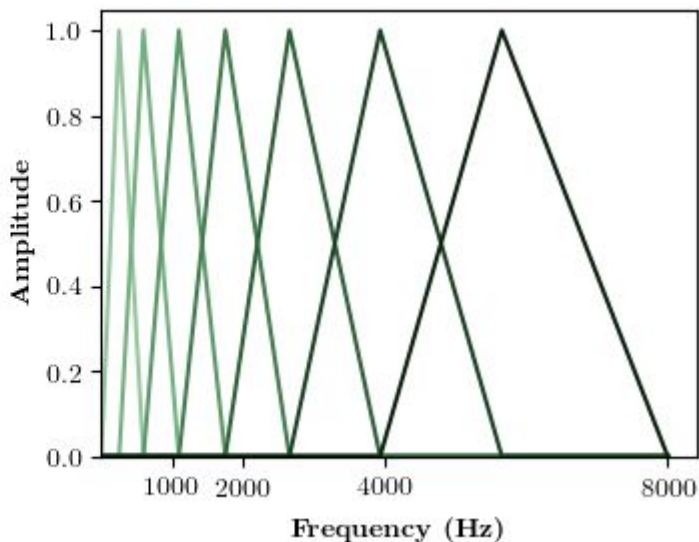
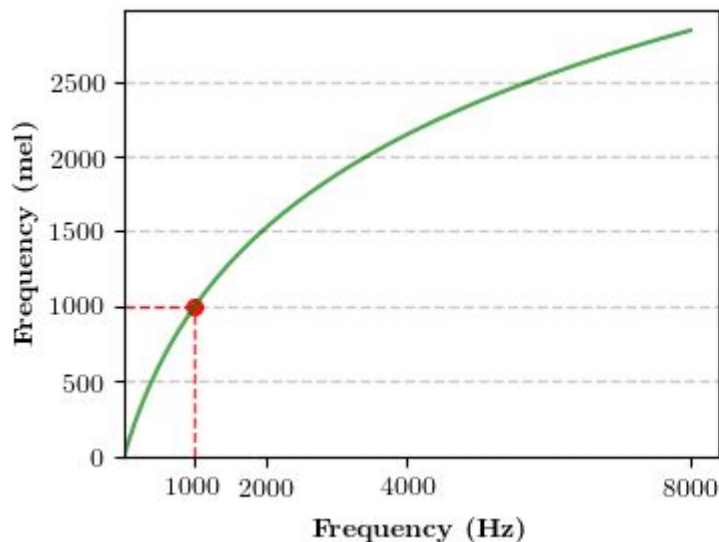
Cochlear Position	Approx. Frequency Range	Speech Relevance
Base	8–20 kHz	Fricatives (/s/, /f/)
Middle	500 Hz – 4 kHz	Most speech intelligibility
Apex	20–500 Hz	Pitch, vowels



Ear does not hear frequencies linearly

- Linear at low frequencies
- Logarithmic at high frequencies
- Instead of uniform Frequency bins, map linear frequencies $\rightarrow$ cochlear positions: Given as **mel-scale**

$$\text{mel}(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$



Sound units in an Indian Language

Characters representing vowel sounds

Short vowels	/a/(अ)	/i/(इ)	/u/(उ)	/e/(ए)	/o/(ओ)
Long vowels	/a:/(आ)	/i:/(ई)	/u:/(ऊ)	/e:/(ऎ)	/o:/(औ)
Diphthongs	/ai/(ऐ)		/au/(औ)		

Characters representing Consonant-Vowel (CV) combinations with vowel /a/(अ)

Place of articulation	Manner of articulation				Nasals	Semivowels	Fricatives
	Unvoiced		Voiced				
	Unaspirated	Aspirated	Unaspirated	Aspirated			
Velar	/ka/(क)	/kha/(ख)	/ga/(ग)	/gha/(घ)	/kna/(ङ)	----	/ha/(ह)
Palatal	/cha/(च)	/chha/(छ)	/ja/(ज)	/jha/(झ)	/chna/(ञ)	/ya/(य)	/sha/(श)
Retroflex	/ṭa/(ट)	/ṭha/(ठ)	/ḍa/(ड)	/ḍha/(ढ)	/ṭna/(ण)	/ra/(र)	/shha/(ष)
Dental	/ta/(त)	/tha/(थ)	/da/(द)	/dha/(ध)	/na/(न)	/la/(ल)	/sa/(स)
Bilabial	/pa/(प)	/pha/(फ)	/ba/(ब)	/bha/(भ)	/ma/(म)	/va/(व)	----

Short time analysis of Speech

- A phoneme (like /a/) doesn't have a single, static acoustic property.
- Its production involves continuous movement of the articulators (tongue, lips, jaw).
- Therefore, the acoustic signal changes *within* the phoneme.

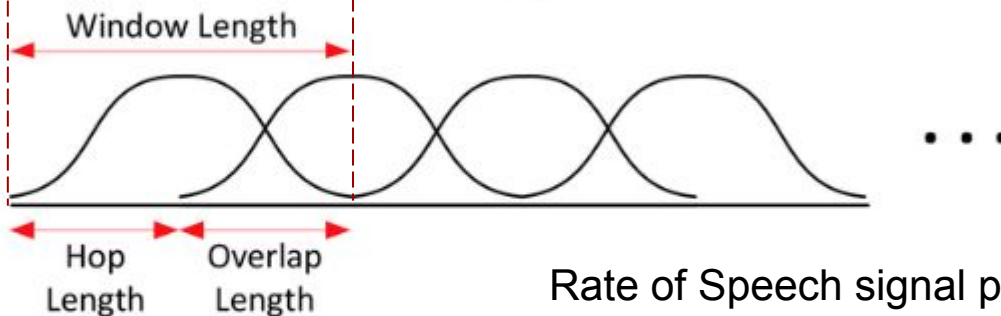
Short time analysis

- The continuous speech waveform is divided into short, overlapping frames (typically 20-25 ms long, with a 10 ms shift between frames).
- Each frame is assumed to be quasi-stationary (statistically stable within that short window).

Signal



Windows



**Segmentation
/ Windowing**



Rate of Speech signal processing

$$\text{Hop Length} = \text{Window Length} \times (1 - \text{Overlap}\%)$$

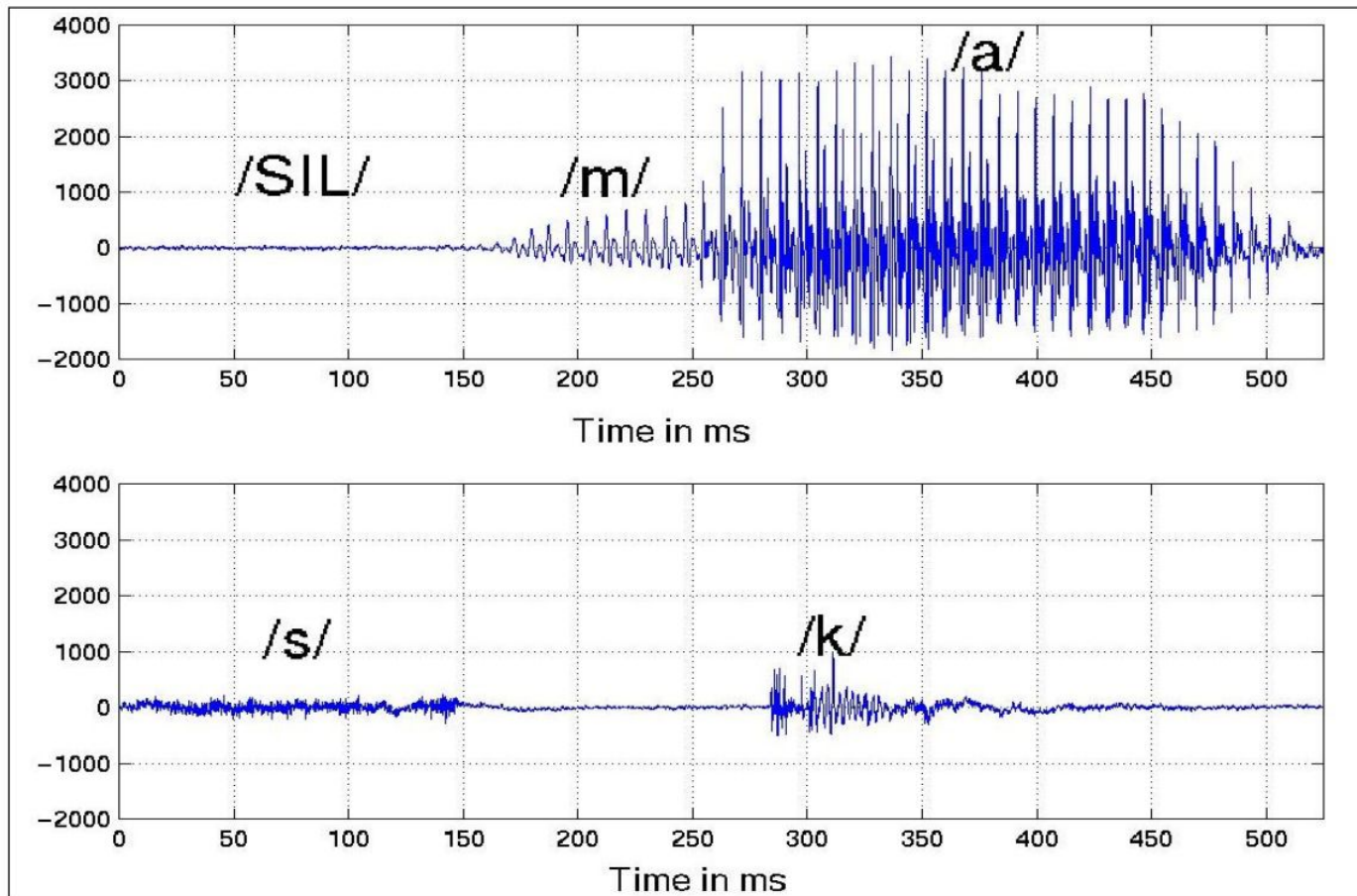
$$\text{Overlap}\% = (1 - \text{Hop Length} / \text{Window Length}) \times 100$$

$$\begin{aligned} \text{Frame Rate} &= \text{Sample Rate} / \text{Hop Length (in samples)} \\ &= 1 / \text{Hop Length (in sec)} \end{aligned}$$

Compute frame rate of a speech signal with $f_s=16\text{KHz}$, 20ms window and 25% frame overlap.

Short time speech analysis

m/a/s/k
n/v/f/stop



Speech Signal Processing

- Constraints of production at various levels in the signal
- Gross level features: Periodic, UV, burst, sil
- Source-System model
- Subsegment level (within a frame $< 40\text{ms}$): frame level features, MFCC
- Segment level (50-200ms): V/UV, Pitch, formants
- Suprasegment level ($>200\text{ms}$ /utterance level): F0 contours, duration, coarticulation
- Time-freq. representation: Temporal, spectral, spectrograms

Energy of a signal

<i>Property</i>	<i>Continuous</i>	<i>Discrete</i>
Energy	$E_x = \int_{-\infty}^{\infty} x(t) ^2 dt < \infty$	$E_x = \sum_{n=-\infty}^{\infty} x[n] ^2$
Power	$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) ^2 dt < \infty$	$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{n=-\infty}^{\infty} x[n] ^2 < \infty$